AIR HOCKEY SIMULATOR

Team 1:

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Contents

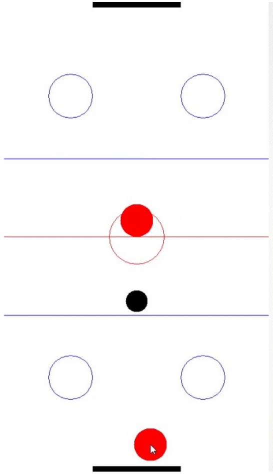
I. Introduction & Demo scenario

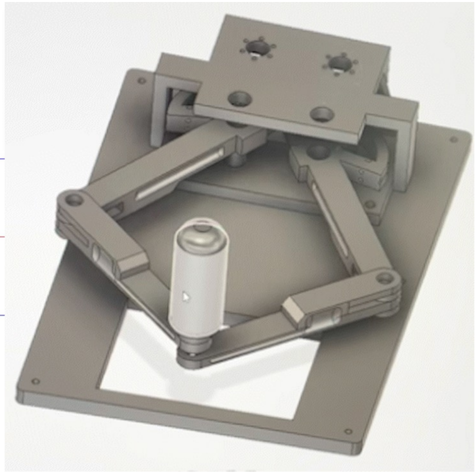
Why Air Hockey?

Background & Motivation

Air hockey is **one of the most natural activity**
to rendering with our **2-DoF kinesthetic haptic device**

- Device limitation
  - Strictly **2D planar workspace**
  - Handle with free rotation: No torque feedback
- Air hockey
  - Relies solely on **2D translational physics**



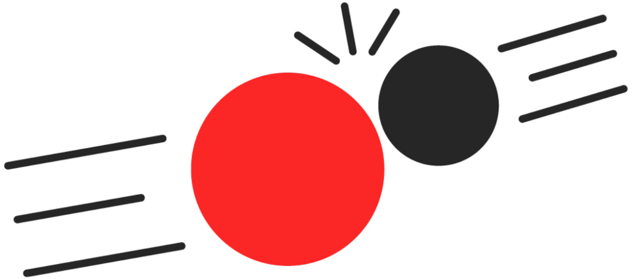


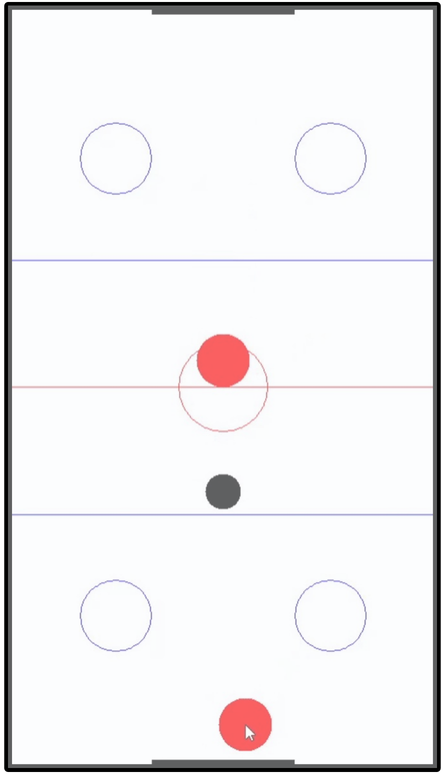
What should be implemented?

Our haptic interaction point
= Mallet (Handle)

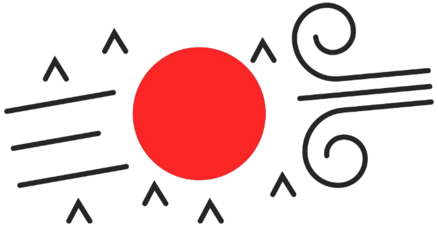
Environment components
= Puck and table edge

- Impact
  Collision between mallet & puck
  Dynamic force rendering

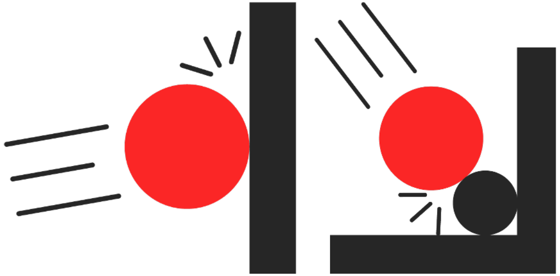




- Friction from air cushion
  Floating ~Various floor condition
  Velocity-based **damping control**



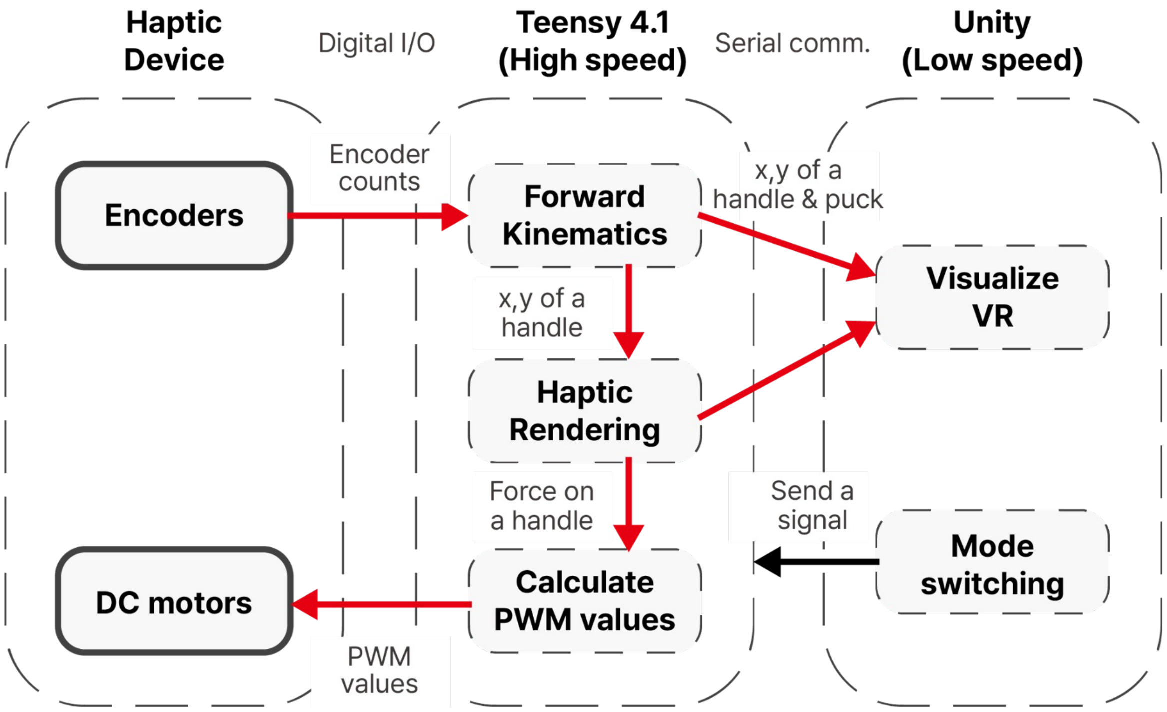
- Virtual wall
  Table edge & puck-squeezing
  High-impedance **boundaries**



Our demo scenario

II. System design

System architecture



- Embedded VR engine
  Teensy **computes all physics**
- No bottleneck
  Unity only **visualizes VR**

→ Small sampling time, Higher Z-width, can render virtual walls better

Hardware Structure Overview

Hardware Improvement for Stable 2-D Haptic Rendering

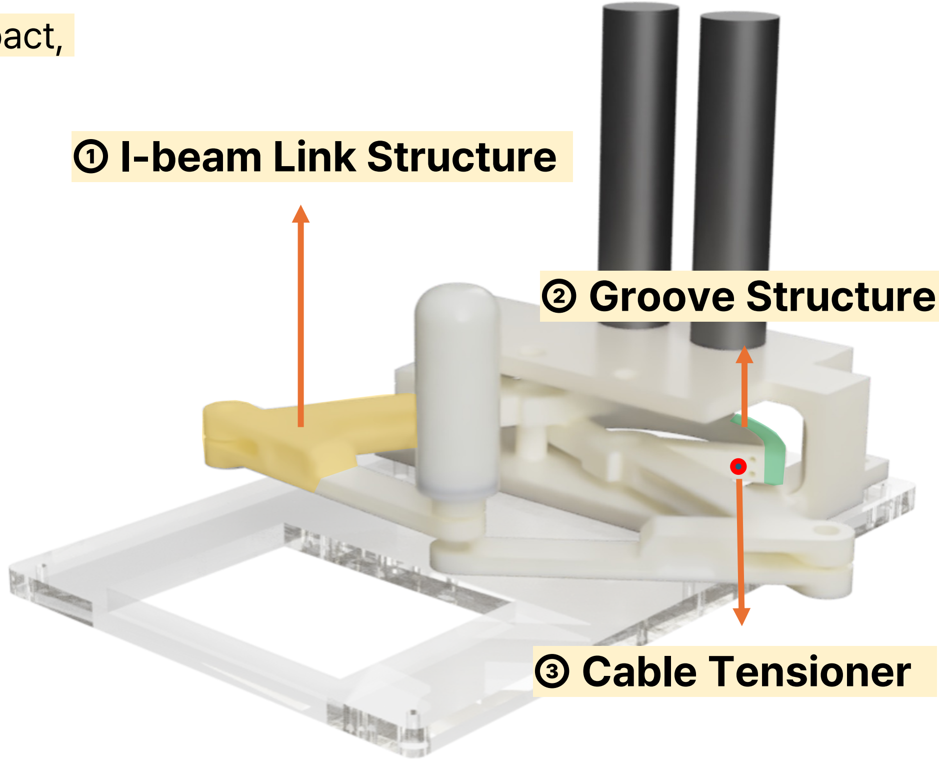
To improve **mechanical stability** for clearer impact, friction and the other haptic feedback

Key Improvements

- Rigid link structure
- Stable cable routing
- Easy cable tension adjustment

Overall Effect

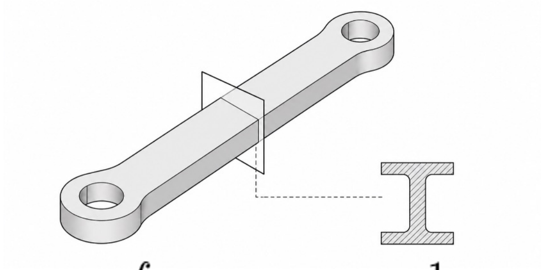
- Smoother planar motion
- More consistent force transmission
- Clearer haptic feedback



Key Mechanical Improvements

① I-beam Link Structure

Improve bending stiffness

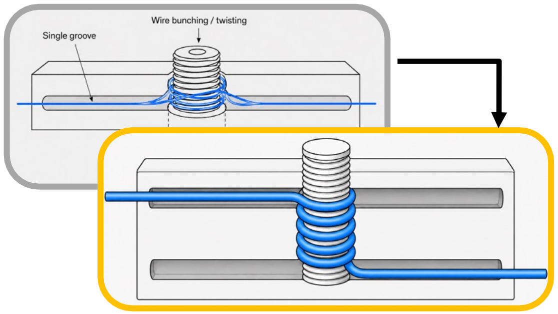


$$I = \int_A y^2 \, dA, \qquad \delta \propto \frac{1}{EI}$$

Increases the **second moment of area**
Reduces **unwanted link deformation**
Supports **consistent force transmission**

② Groove Structure

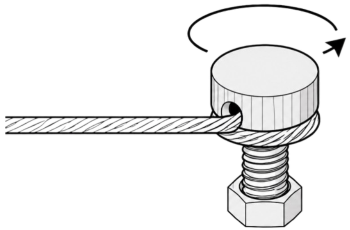
Guide cable routing



Prevents the cable from **slipping** out of its path
Double grooves guide upper & lower cable paths
Compensates cable height change on capstan
Reduces cable-link interference

③ Cable Tensioner

Adjust cable tension easily



Bolt-based **tension adjustment**
Makes **calibration and maintenance** easier
Improves **repeatability** of haptic feedback

III. Haptic Rendering

Haptic Rendering

Unity UI Configuration

1. Score Tracking

Real-time visual feedback of gameplay progress.

2. Reset Function

Initialize the position of the puck and handle.

3. Puck Weight Slider

Adjustable slider to modify haptic resistance & impact.

4. Ground Material

Multi-option selection for different surface textures.

Haptic Rendering

Total Rendering Force

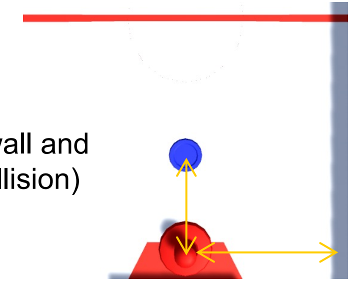
$$F = F_{wall} + F_{collision} + F_{damping} + F_{texture}$$

$$= \boxed{-k_{wall} x_{penetration}} \boxed{-(k_{puck} m_{puck}) d\hat{n}} \boxed{-b_{floor} v} \boxed{+A_{vib} \sin(2\pi f_{spatial} x)}$$

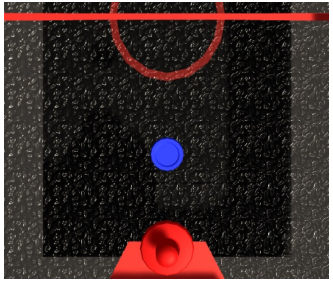
Wall Spring **Collision Spring** **Damping** **Vibration**

Default terms

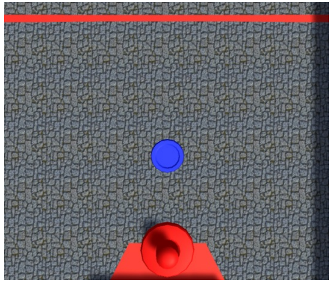
(Consider the wall and mallet-puck collision)



For a friction ground



For a fine-grained ground



Haptic Rendering

① Penetration-based spring model (Wall & Collision)

Enable only basic collision and wall rendering

$$F = F_{wall} + F_{collision}$$
$$= -k_{wall}x_{penetration} - (k_{puck}m_{puck})d\hat{n}$$

$$d = (r_{mallet} + r_{puck}) - \|x_{puck} - x_{handle}\| \ , \quad \hat{n} = \frac{x_{puck} - x_{handle}}{\|x_{puck} - x_{handle}\|}$$

- k_{wall}: wall stiffness
- $x_{penetration}$: penetration depth into the wall

- k_{puck}: base collision stiffness
- m_{puck}: puck mass
- d: penetration depth between mallet and puck
- $\hat{n}$: contact normal direction

- r_{mallet}: radius of the haptic mallet
- r_{puck}: radius of the virtual puck
- x_{puck}: current position of the virtual puck
- x_{handle}: current position of the haptic handle

② Velocity-based friction (Damping)

Add damping friction

$$F = F_{wall} + F_{collision} + \boxed{F_{damping}}$$
$$= -k_{wall}x_{penetration} - (k_{puck}m_{puck})d\hat{n} \boxed{- b_{floor}v}$$

- b_{floor}: viscous damping coefficient
- v: handle velocity

③ Vibration-based texture (Vibration)

Add spatial texture rendering

$$F = F_{wall} + F_{collision} + \boxed{F_{texture}}$$
$$= -k_{wall}x_{penetration} - (k_{puck}m_{puck})d\hat{n} \boxed{+ A_{vib}\sin(2\pi f_{spatial}x)}$$

- A_{vib}: vibration amplitude
- $f_{spatial}$: spatial frequency of the texture field
- x: current handle position

IV. Conclusion & Future Work

Conclusion & Future Work

Conclusion

- Built a 2-D planar haptic system for **hockey-like interaction**
- Rendered **impact, friction** and **vibration** through haptic feedback
- Integrated hardware, control logic, and haptic rendering
- Achieved stable wire routing/tension while maintaining smooth user motion

Future Work

- Reduce friction, slack, and backlash
- Extend to teleoperation with haptic feedback

Thank you